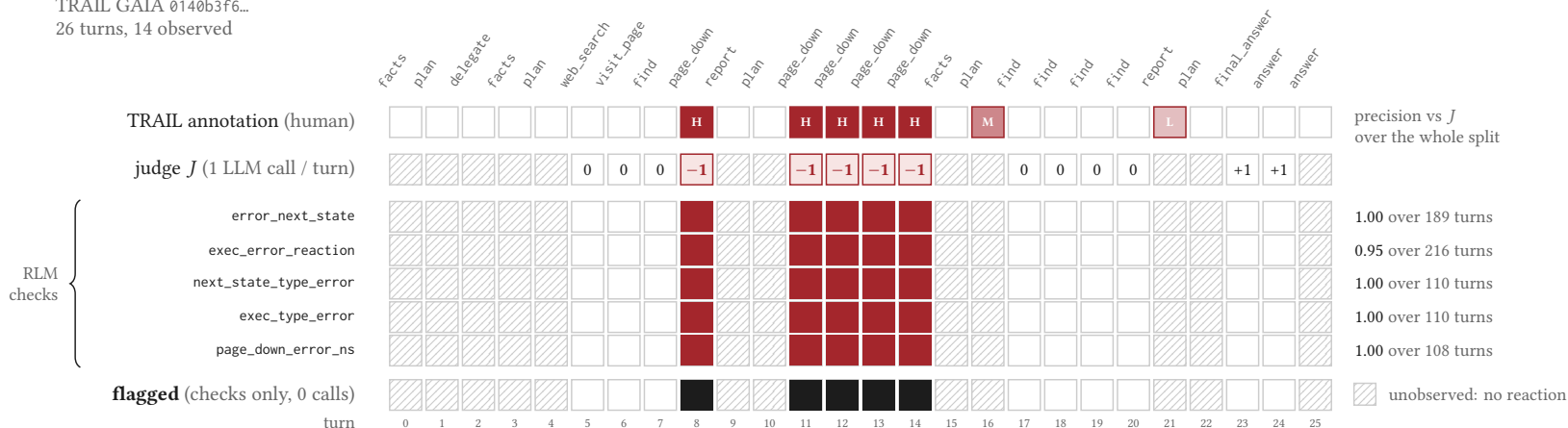


One real trace, turn by turn

TRAIL GAIA 0140b3f6...

26 turns, 14 observed



turn 7 $J = 0$, no check fires

a_t find_on_page_ctrl_f("equine veterinarian")
s_t+1 The search string 'equine veterinarian' was not found on this page.

hint: "the agent could try alternative search terms."
A miss, not a mistake. Annotators marked nothing here either.

turn 8 $J = -1$, all five checks fire

a_t page_down({"": "", "arguments": {}})
s_t+1 AgentExecutionError: ... TypeError:
PageDownTool.forward() got an unexpected keyword argument ''

Annotated: Formatting Error, HIGH. The agent repeats the mistake on turns 11–14; every repeat is caught.

turns 16, 21 unobserved, never scored

A planning step and a hand-off report. Annotators marked a Context Handling error (MED) and a Language-only error (LOW).

Neither has a reaction to read, so neither can be caught here. That is the method's stated limit, not a silent miss.

```
// export-nextstate row (abridged)
{"label": "negative", "flagged_turns": [8, 11, 12, 13, 14], "flagged_by": {"8": ["error_next_state", "exec_error_reaction", ...], ...},
"checks_applied": [5 checks], "all_checks_reviewed": false, "verifier_id": "family-verifier-dadbd15ac3d641", "judge_run_id": "turn-judge-0044b8f5c041155c", "messages": [...]}
```

score = $1 - 5/14 = 0.64$; fails. all_checks_reviewed is false because this run applied --survivors rather than a human review.

Figure 2. A worked example. One TRAIL GAIA trace, end to end. Columns are turns, and hatched columns have no reaction. The judge scores the fourteen observed turns. The five checks the RLM wrote, each re-verified on the whole split (right), fire on exactly the turns the judge scored -1 . Those are the five turns that TRAIL's annotators independently marked as HIGH-impact errors. Scoring with the checks alone takes no model call and reproduces the judge's flags here. The two annotated errors the pipeline does not catch sit on turns without a reaction, which the method does not score by design. All values are read from the saved artifacts.